

Questions 2022

Q: Can you tell me something about Article 22 of the GDPR about the explainability?

A: The data subject shall have the right not to be subject to a decision based solely on automate processing, including profiling, which produces legal effects (law enforcement) concerning him or her or similarly significantly affects him or her (creditworthiness assessment).

Exceptions

- a. Necessary for entering into, or performance of, a contract between the data subject and a data controller. (e.g, CV evaluation of a really high number of applications in a inhuman amount of time)
- b. Authorised by the Union or Member State law to which the controller is subject and which also lays down suitable measures to safeguard the data subject's rights, freedoms and legitimate interests
- c. Based on the data subject's explicit consent.

Conditions for application

1. A decision must be taken
2. It must be solely based on automated processing
3. It must include profiling (your Bancomat blocking your card after enough trials of the pin is an automated decision but not based on profiling)
4. It must have legal or anyway significant effect.

Information on automated decision making

- Input data and whether different data items are favouring or disfavouring the outcome.
- Target values
- Possible consequences of the decision
- Overall purposes of the system

Right to explanation (THE ONE SPECIFICALLY ASKED ON THIS QUESTION)

- Right to obtain human intervention

- Right to express data subject's POV
- Right to challenge the decision

Missing on Article 22 but mentioned on Recital 71

- Specific information
- Right to obtain an explanation of the decisions reached after such assessment

There are two interpretations for this missing; i) either there is no legal enforcement but rather a recommendation, or ii) there is legal enforcement whenever technology and costs allows it

Q: What are the legal basis for the processing of personal data as stated in the GDPR?

A: First of all, let's define the *personal data*. Personal data means any information relating to an identified or identifiable natural person (data subject), who can be identified, directly or indirectly, in particular by reference to an identifier such as name, id number, location data, etc...

Processing of this *personal data* shall be lawful only if and to the extent that one of the following applies (Article 6):

- a. The data subject has given direct consent
- b. The processing is necessary for the performance of a contract
- c. The processing is necessary for the compliance with legal obligation
- d. The processing is necessary in order to protect vital interests
- e. The processing is necessary for the performance of a task carried out in the public interest
- f. The processing is necessary for the purposes of the legitimate interests pursued by the controller or a third party. **Except!** ⇒ where such interests are overridden by the interests of fundamental rights and freedoms of the data subject.

Q: Can you tell me some of the issues related to filtering? How is it regulated?

An example of this could be the increment on difficult to escape from filter bubbles or confirmation bias. If a recommendation systems always filters based in our tastes we might end up having a flat opinion. Another issue could

be behavior influence or manipulation, through filtering information only related to one kind of ideology we could slightly change the ideology of a person. This also has consequences on fundamental rights like freedom of speech.

Regarding legal measures we have the Article 21 of the GDPR where it talks about objecting to profiling ⇒ The data subject has the right to object where personal data are processed for direct marketing purposes at any time to processing of personal data concerning him or her, which includes profiling to the extent that it is related to such direct marketing.

The controller has the obligation to provide; I) information on the existence of automated decision-making, II) the logic involved as well as the consequences of such processing.

Q: Can you tell me something about the ethical issues related to Autonomous Vehicles? About the MIT Lab experiment, is there an approach that you find more fair than other?

The main issues related to AVs are the following:

- Disputes of liability. Who is legally responsible in case of an accident?
- Loss of privacy and security. Provider knows where you are going all the time and the AVs could be hacked
- Resistance to loose control (?)
- Legal issues and ethical dilemmas. In case of unavoidable accident, who'd you kill?

About the Moral Machine project from the MIT Lab, for me the more fair approach would be to comply with traffic regulations as much as possible as keeping the life of the persons on board the safest possible. Let's imagine an unavoidable accident, the idea should be to evaluate the risks for the passengers, if this fails, let's say we cannot guarantee survival of the passengers (we could still give up some harm on the passengers to avoid a fatal accident on a third party) then we should sadly comply to traffic regulations. We could implement a kind of utilitarian approach that takes into account the risks for the passengers. Why is this? Cause with this approach we could always be legally justified even though we could argue about the morality.

Q: Can you tell me something about the idea of utilitarianism? What are the problems related to wealth distribution?

It is a school of thought based on consequentialism, this is, the morality of actions is based solely on the consequences of that action. One of the most common approaches to this school of thought is the utilitarianism, which basis the morality of an action in the outcome that it gives known as utility. The utility of an action can be defined in different ways, but usually is the difference between the happiness, pleasure or absence of pain and the unhappiness, pain or privation of pleasure.

The main advantages of utilitarianism are:

- Conceptually simple
- Egalitarian (Everyone is equally important)
- Fits with basic intuitions

However, some issues are:

- What about relatives and friends? We cannot give priority?
- To what extent should I give my goods to others? To the minimum I can survive? Is this fair for me? ⇒ Wealth distribution problem

Regarding the wealth distribution some issues are:

- Is it ok to make an action that benefits some to the detriment of others?
- Is it fair to sacrifice all my wealth for others?

Some economists adopt the wealth maximisation approach, which aims at maximising the wealth in society regardless of distribution.

Q: Can you tell me something this idea of fairness in automated decision making, what are the main issues related to this idea of fairness?

In some cases algorithms engage in explicit unlawful discrimination, called disparate treatment basing their outcome on prohibited features such as race ethnicity or gender (direct discrimination). More often a system's outcome will be discriminatory due to its disparate impact (indirect discrimination).

Sources of AI unfairness are:

- System's architecture design is biased (rare)

Direct discrimination sources

- Dataset based on sensitive predictors
- **Learning of past (and biases) human decisions**

Indirect discrimination sources

- Dataset not representing population (biased)
 - Missrepresenting behavior of a group
 - Underrepresenting a group
- Tracking of non-discriminatory goal through discriminatory proxy

Q: Let's say we have a system that is equally accurate for two groups (blue and green). If you have the same features in both groups you will be treated equally. Then it may happen that the kind of error made by the system affect different to the two groups. Being considered false negative is better than being considered false positive. Can you make an example?

A: This is an example of unfairness in the decision, for example, for law enforcement, a false negative is much worse than a false positive. You could be wrongly set in jail due to a false negative error.

Q: Do you remember the principle of data minimisation? Is there a tension between this and AI methods? The GDPR provides some ways for re-using the data, do you know these ways, it means, exceptions by which you can re-purpose the data.

A: As stated by **Article 25** of the GDPR, the controller shall implement appropriate technical and organisational measures, such as pseudonymisation for data-protection and data minimisation to protect the rights of data subjects at time of determining the means for processing and at time of processing itself.

Regarding AI methods, yes, this is a problem since the more data you collect for a machine learning model, the better will be the accuracy. However, we have to be aware of the importance of data privacy and protect the data subjects according to the GDPR.

Regarding the right to erasure from Article 17, there are some exceptions by which the controller can keep the data if the processing is necessary for:

- a. exercising the right of freedom of expression and information
 - b. compliance with a legal obligation
 - c. the public interest (the area of health or research purposes)
 - d. exercise or defense of legal claims
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Q: Do you remember Deontology?

A: Deontology is a school of thought in normative ethics which states that certain actions are good or bad regardless of their consequences. The right has priority over the good, what makes a choice right is its conformity with a moral norm which order or permits it, rather than its good or bad effect.

Golden rule ⇒ Treat others as you would like other to treat you. Do not treat others in ways that you would not like to be treated.

One approach to deontology school of thought is the Kantian ethics where he defines the **categorical imperative**, this is a moral rule that everyone must follow, no matter what they want or need. The categorical imperative is a duty that comes from reason, meaning that being moral is part of being rational. Unlike this, there are "hypothetical imperatives," which tell us what to do to achieve our specific goals.

In order to understand if an imperative is categorical or hypothetical, Kant proposes four different formulas, the most known is the concept of universalizability. Universalizability says to act only according to a maxim by which you can at the same time will that it should become a universal law

A maxim is a subjective principle of action ⇒ Connects an action to the reasons for it.

- Should one must tell a known murderer the location of his prey?
 - For Kant, the maxim of telling lies is not universalizable

The universalisation test by Shafer Landau is a test to check the universalizability of an action. This is done by formulating your maxim clearly, then imagine a world in which everyone supports and acts on your maxim. Can the goal of my action be achieved in such a world?

Q: Do you remember the theory of justice by John Rawls?

A: The justice theory of John Rawls is a theory inside social contract theories which states that an action is morally right just because it is permitted by rules that free, equal, and rational people would agree to live by, on the condition that others obey these rules as well. As said by John Rawls, this people should agree by choosing under a **veil of ignorance**, this is regardless gender, social position, interests, talents, wealth, race, etc...

There are two principles pursued by this theory:

1. Each person has the same infeasible claim to a fully adequate scheme of equal basic liberties, which is the same for everyone.
 2. Social and economic inequalities are to satisfy two conditions
 - a. They have to be attached to offices and positions open to all under conditions of fair equality opportunity.
 - b. They have to be the greatest benefit of the least-advantaged members of society.
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Q: What are the principles of trustworthy AI? What about the accountability?

- Respect for human autonomy
- Prevention of harm
- Fairness
- Explicability

Regarding accountability, accountability is the idea of deciding who is responsible when AI goes wrong. For trustworthy AI, mechanisms for accountability need to be implemented, for example, in the case of AVs, the higher is the level of autonomy, the higher the liability relies on the provider and the less on the driver.

Q: What can you tell me about the idea of human rights?

Human rights are ethical demands concerning freedoms, opportunities including liberty and social rights. They are inherently entitled simply because you are a human being. They are the same for everyone and are applicable everywhere regardless of age, ethnicity, religion, gender, etc...

Some examples are the freedom to speech, right to privacy or the presumption of innocence.

Some times AI can interfere with human rights and that is why we need to contribute for a trustworthy AI that respects human rights.

An example of this is how AI can be used to manipulate and influence human behaviors interfering in this way with the right of free association and liberty.

Q: Do you remember something about the virtue ethics? What are the main issues of this school of thought?

Virtue ethics is a school of thought which states that an act is morally right just because it is one that a virtuous person, acting in character, would do in that situation. A virtuous person is the one that possesses a mix of relevant virtues like honesty, loyalty, courage, impartiality, wisdom, fidelity, generosity, ...

It states that ethics cannot be learned through a set of rules, its application requires practical wisdom.

Some issues related to this ethics are:

- How to know what is a virtue and what is not?
- What if virtues are in conflict? for example pride and humble
- How much can we rely in traditions? Maybe now courage is something and in a few years this can change.

Q: Do you remember the duties of information of a service provider that should be given to a data subject in the context of collection and processing of personal data under the GDPR? What are the rights that the data subject can exercise? And the rights for data access?

- Identify of the controller
- Contact details of the data protection officer
- Purposes of the processing for which the personal data are intended
- Legal basis for the processing
- Categories of personal data concerned
- Recipients or categories of recipients
- Period for which the personal data will be stored
- Existence of the right to request from the controller access to and rectification or erasure of personal data. Existence of the right to data portability.
- Right to lodge a complaint
- Source from which personal data originate
- Existence of automated decision-making

The rights that a data subject can exercise regarding this question are:

- Right of erasure

- Right to not be subject to a decision based solely on automated processing.
- Right to object to profiling and direct marketing
- Right to withdraw the consent at any time

The rights that a data subject can exercise regarding the access of data are:

- Right to access to both, personal data and inferred output.
- Right to rectification

Q: Do you remember the Cambridge Analytica case?

It was a case exhibiting the dangers of profiling. They basically built a database of voters in the USA by inviting them to take a political test in exchange of money. The voters were also asked to give access to their facebook pages, allowing to connect the answers in the test to the information on the facebook page of each voter. Cambridge Analytica, not only retrieved the information of the voters, but also the one belonging to their friends. With this two sets of data, Cambridge Analytica was able to train a model based on the first one to infer political and psychological ideas of the retrieved profiles on the second set.

Based on this profiling, voters who were likely to change their voting behavior were identified, if provided appropriate messages and political advertisement they could be manipulated or influenced without them being aware of it.